What Does the Model Learn About the Curvature? An Empirical Study of Learned Poincaré Curvature in Hyperbolic Zero-Shot Learning

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Abstract

A prior result on AwA2 Bejleri [2025] reported that the learnable curvature c of a Poincaré ball hyperbolic zero-shot learning model scaled monotonically with training data, suggesting a universal $c \propto \log N$ relationship. We test this on seven image classification datasets spanning attribute-based and text-based class embeddings, and from very flat (Stanford Cars) to deeply hierarchical (iNaturalist, tiered-ImageNet) label spaces. We find that the relationship is *not* universal. The within-dataset slope of c vs. $\log N$ *changes sign* across our seven datasets, from $+1.31$ (AwA2, $R^2 = 0.95$) to -0.17 (SUN397, $R^2 = 0.86$), ruling out a universal positive scaling law. All results are from a single training seed per (dataset, ablation) cell; quantitative claims about slope magnitudes should be read with that caveat. Instead, we identify three distinct response patterns: (1) *strongly increasing* curvature with data (AwA2); (2) *rising then plateauing* curvature (CUB-200, Stanford Cars, CIFAR-100); (3) *peaking early then declining* curvature (SUN397, iNaturalist, tiered-ImageNet). The single biggest determinant of which pattern a dataset exhibits is the *type* of class embedding used, not its taxonomy depth: AwA2 uses 85-d expert-annotated binary attributes (e.g., `has_stripes`, `carnivorous`), while every other dataset uses 300-d GloVe word vectors derived from class names. We argue that learned curvature reports the dataset's *available semantic structure* as seen through the chosen embedding, not its underlying taxonomy depth. This finding has consequences for the recent generalisation results of Fan et al. Fan et al. [2025]: their reported optimal curvature should be interpreted relative to the embedding modality, not as a property of the dataset's classes per se.

1 Introduction

Hyperbolic spaces are natural targets for embedding hierarchical data Nickel and Kiela [2017, 2018], and modern hyperbolic neural networks Ganea et al. [2018], Bécigneul and Ganea [2019], Kochurov et al. [2020] commonly treat the Poincaré ball curvature c as a learnable parameter rather than a fixed hyperparameter. A natural question is what the model is doing when it learns this parameter — in particular, what information about the dataset is encoded in the converged value.

In an earlier paper, we trained a Bayesian hyperbolic zero-shot learning (ZSL) model on AwA2 across a data-ablation sweep and observed that c scaled monotonically with training set size, from $c = 1.18$ at 1% data to $c = 6.63$ at 100% Bejleri [2025]. The relationship looked clean enough on a log scale to invite a strong reading: that learnable curvature obeys a universal $c \propto \log N$ scaling law, with the slope and intercept perhaps controlled by hierarchy depth. This paper tests that reading.

What we did. We trained the same model architecture — with no hyperparameter changes — on seven image classification datasets covering a wide range of taxonomic structures and class-embedding modalities: AwA2 Xian et al. [2019], CIFAR-100 Krizhevsky [2009], CUB-200 Wah et al. [2011], SUN397 Xiao et al. [2010], Stanford Cars Krause et al. [2013], iNaturalist Van Horn et al. [2021], and tiered-ImageNet Ren et al. [2018]. Each was run through the same data-ablation sweep at $\{1, 5, 10, 25, 50, 100\}\%$ of the training set.

What we found. The $c \propto \log N$ pattern does *not* generalise. Fitting $c = a \log N + b$ separately per dataset, the slope a varies from $+1.31$ (AwA2) to -0.17 (SUN397), changing sign across the seven datasets. The within-dataset trajectories cluster into three qualitatively distinct response patterns (Section 5.2):

1. **Monotonic increase** — AwA2 is the only dataset in this group, climbing 5.6$\times$ from 1% to 100% data.
2. **Rise then plateau** — CUB-200, Stanford Cars, and (loosely) CIFAR-100. Curvature rises to a moderate ceiling around $c \in [1.6, 2.3]$ by 25% data, then stays flat or drifts down slightly.
3. **Peak then decline** — SUN397, iNaturalist, and tiered-ImageNet. Curvature peaks early (often at 5–10% data) and then *decreases*, sometimes ending below its initial value.

The taxonomy-depth ordering of the datasets does not predict which group a dataset falls into. The strongest predictor we identify is the *class-embedding modality*. AwA2 is the only dataset using expert-annotated per-class binary attributes (85-d, e.g., `has_stripes`, `carnivorous`); the other six use 300-d GloVe word vectors Pennington et al. [2014] derived from class names. The 4$\times$ gap between AwA2's converged curvature ($c = 6.63$) and the next-highest dataset (CIFAR-100, $c = 1.81$) lines up with this distinction far more cleanly than with anything taxonomic.

What it means. We re-frame the original question. Rather than asking "what does converged curvature reveal about the dataset?", the right question appears to be: "what does converged curvature reveal about the dataset *as seen through its class-embedding modality?*" A model that has access to rich, structured semantic features (AwA2's attributes) finds increasing geometric value in additional data and continues compressing into more curved space. A model that sees classes only as text-derived word vectors hits a curvature ceiling well below this and, on visually rich datasets with many classes, even unlearns its initial curvature once enough data is available. In Fan et al.'s concurrent work Fan et al. [2025], an "optimal" curvature is derived for individual datasets via PAC-Bayesian bounds; our results suggest those values are best understood as functions of the (dataset, embedding modality) pair rather than of the dataset alone.

Contributions.

1. We refute the claim that learnable curvature in hyperbolic neural networks obeys a universal $c \propto \log N$ scaling law, using seven datasets in a controlled experimental setup that isolates dataset and ablation as the only varying factors.
2. We identify three reproducible response patterns and characterise each.
3. We provide evidence that class-embedding modality (binary attributes vs. word vectors) is a stronger predictor of curvature dynamics than taxonomy depth, and discuss the implication for prescriptive curvature-tuning work.

2 Related Work

Hyperbolic representation learning. Nickel & Kiela Nickel and Kiela [2017] demonstrated that hyperbolic embeddings can capture taxonomic structure with far fewer dimensions than Euclidean embeddings. Ganea et al. Ganea et al. [2018] extended hyperbolic representations into deep networks by defining hyperbolic linear layers, attention, and learnable curvature. Subsequent work introduced Riemannian optimisers Bécigneul and Ganea [2019] and tooling Kochurov et al. [2020]. The Wrapped Normal distribution on hyperbolic space Mathieu et al. [2019], Nagano et al. [2019] provides the basis for variational hyperbolic models, including the ZSL framework we use Bejleri [2025].

Curvature learning and generalisation. Fan et al. Fan et al. [2025] derive PAC-Bayesian generalisation bounds for hyperbolic networks that depend on curvature, showing that curvature affects loss-landscape sharpness, and propose a sharpness-aware bi-level optimisation procedure to learn curvature for better generalisation. They report converged curvature values on tiered-ImageNet ($c \approx 0.113$ for 1-shot, $c \approx 0.127$ for 5-shot) and CIFAR-10-LT ($c \approx 5.36 \times 10^{-4}$). Their contribution is **prescriptive**: how to set curvature for performance. Ours is **descriptive and diagnostic**: what does the curvature the model learns tell us about the dataset and its embedding? The two are complementary in principle, but our results suggest that the same dataset can produce qualitatively different curvature behaviour under different class embeddings, and the comparison must be done carefully.

Hierarchy and δ-hyperbolicity. Gromov Gromov [1987] introduced the four-point condition for measuring how tree-like a metric space is, with $\delta = 0$ corresponding to a tree. Fournier et al. Fournier et al. [2015] provide efficient algorithms for computing δ. We use taxonomy-derived hierarchy metrics (depth, branching factor) as our primary structural descriptors and discuss why they correlate poorly with the learned curvature in our setup.

Table 1: The seven datasets used in this study, with class-embedding modality and ground-truth taxonomy metrics. Hierarchy depths and branching factors are computed from the simplified taxonomies in our `hierarchy.py`: for tiered-ImageNet (full WordNet subtree depth $\approx$ 12–15) and iNaturalist (full biological taxonomy depth 7) these are conservative truncations to the most coarsely meaningful levels (kingdom/phylum/class/order for iNat, the major superclass split for tiered-ImageNet). The truncation tightens the depth range across our datasets but does not change the qualitative ordering — iNaturalist and tiered-ImageNet remain the deepest — so its effect is to weaken any depth-vs-curvature signal slightly, which biases against rather than for our finding that depth is a poor predictor.

Dataset	**Classes**	**Embedding**	**Embed dim**	**Tree depth**	**Branching**
AwA2 Xian et al. [2019]	50	per-class attrs	85	3	2.69
CIFAR-100 Krizhevsky [2009]	100	GloVe	300	2	5.71
CUB-200 Wah et al. [2011]	200	GloVe	300	3	7.83
SUN397 Xiao et al. [2010]	397	GloVe	300	3	7.24
Stanford Cars Krause et al. [2013]	196	GloVe	300	2	10.50
iNaturalist Van Horn et al. [2021]	500	GloVe (taxonomy)	300	4	4.62
tiered-ImageNet Ren et al. [2018]	608	GloVe (WordNet)	300	5	4.10

3 Method

We use the Bayesian Hyperbolic ZSL architecture from Bejleri [2025] with no modifications. A visual encoder maps pre-extracted ResNet-101 features He et al. [2016] (2048-d) to a Wrapped Normal distribution $\mathrm{WN}(\mu, \sigma^2)$ on the Poincaré ball $\mathbb{B}_c^d$ Nagano et al. [2019]. Throughout this paper we follow the `geoopt` Kochurov et al. [2020] parameterisation: the ball $\mathbb{B}_c^d$ has *constant negative sectional curvature* $-c$ for $c > 0$, so larger c means a more strongly hyperbolic geometry (sharper exponential growth of ball volumes near the boundary), and $c \to 0$ recovers the Euclidean limit. Whenever we write "higher curvature" we mean larger c in this convention. The mean μ is produced by a hyperbolic linear layer Ganea et al. [2018] and the log-variance $\log \sigma^2$ by a standard linear head. A semantic encoder maps class-level embeddings to deterministic points on the same manifold. Training minimises a geodesic alignment loss plus a KL regulariser ($\lambda_{\mathrm{KL}} = 0.01$) using RiemannianAdam Bécigneul and Ganea [2019].

Hyperparameters held fixed across all datasets. Hidden dim 1024; embedding dim 64; learning rate 10^{-3} for Euclidean parameters and 10^{-4} for hyperbolic curvature; temperature 0.1; initial curvature $c_0 = 1.0$; 200 epochs; batch size 128 (clamped to $\min(\text{batch size}, n_{\text{train}})$ for very small ablations to avoid empty data loaders). This ensures any cross-dataset differences are attributable to the dataset itself and its class embeddings, not training configuration.

4 Datasets

Table 1 lists the seven datasets and their key properties. The two columns most relevant for our analysis are the **embedding modality** and the **tree depth**. Among our seven datasets, only AwA2 uses expert-annotated per-class binary attributes; the other six use GloVe word vectors derived from class names (with WordNet lookup for tiered-ImageNet's wnids and a taxonomy-name parse for iNaturalist's directory-encoded names).

We deliberately span a range of label-space structures: shallow non-taxonomic domains (Stanford Cars region/make groupings, SUN397 scene categories), exact taxonomies (CIFAR-100's 20-superclass partitioning), biological/ornithological hierarchies (CUB-200, iNaturalist), the deep WordNet subtrees of tiered-ImageNet, and the AwA2 attribute-based animal description. For each dataset we use pre-extracted ResNet-101 features on the full image set and run the data-ablation sweep at $\{1, 5, 10, 25, 50, 100\}\%$ of the seen-class training data, using the same random seed across datasets for the ablation subsample.

5 Results

5.1 The $c \propto \log N$ scaling claim is decisively rejected

Table 2 reports the converged curvature for each (dataset, ablation) cell. Figure 1 plots the same data on a $\log N$ x-axis.

Fitting a one-line linear model $c = a \log N + b$ separately per dataset gives the slopes in Table 3, including 95% confidence intervals and p-values from the standard t-test against $a = 0$ (Wald, with $n - 2 = 4$ residual degrees of

Table 2: Converged curvature c at epoch 200 across all seven datasets and data-ablation fractions. AwA2's monotonic increase is the dramatic outlier; the GloVe-embedded datasets show much more modest curvature, with two of them ending at or near (and one well below) their initial $c_0 = 1.0$ at full data.

	Training data fraction					
Dataset	**1%**	**5%**	**10%**	**25%**	**50%**	**100%**
AwA2	1.18	2.11	3.24	5.36	6.09	**6.63**
CUB-200	1.06	1.24	1.54	2.14	2.12	**2.01**
CIFAR-100	1.30	2.29	2.22	2.25	1.99	**1.81**
Stanford Cars	1.08	1.17	1.32	1.69	1.70	**1.58**
tiered-ImageNet	1.81	1.77	1.50	1.26	1.27	**1.39**
SUN397	1.58	1.59	1.45	1.23	0.98	**0.85**
iNaturalist	1.07	1.54	1.60	1.37	1.09	**1.08**

Table 3: Per-dataset fit of $c = a \log N + b$ with analytical 95% confidence interval on the slope and p-value vs. $a = 0$. The slope changes sign across datasets, two slopes are statistically indistinguishable from zero, and AwA2's CI does not overlap any other dataset's — jointly falsifying any universal positive scaling claim.

Dataset	**Slope a (95% CI)**	R^2	p
AwA2	$+1.31$ $[+0.90, +1.72]$	0.95	0.001
CUB-200	$+0.26$ $[+0.10, +0.42]$	0.84	0.010
Stanford Cars	$+0.15$ $[+0.05, +0.25]$	0.80	0.016
CIFAR-100	$+0.09$ $[-0.20, +0.38]$	0.15	0.446
iNaturalist	-0.03 $[-0.23, +0.17]$	0.04	0.715
tiered-ImageNet	-0.13 $[-0.23, -0.03]$	0.75	0.026
SUN397	-0.17 $[-0.27, -0.08]$	0.86	0.008

freedom per fit). The signs and magnitudes of a disagree across datasets in three statistically meaningful ways: (i) AwA2's slope is an order of magnitude larger than the next-largest positive (CUB-200) and its 95% CI does not overlap CUB-200's, ruling out a single underlying scaling constant for the two; (ii) two datasets (SUN397, tiered-ImageNet) have slopes whose 95% CI lies *entirely below zero*, so they are not just smaller versions of AwA2's positive slope but qualitatively different decreasing trends; (iii) two datasets (CIFAR-100 and iNaturalist) have slopes whose 95% CI *straddles zero* ($p = 0.45$ and $p = 0.72$), meaning we cannot statistically distinguish them from *no* log-N trend at all. A single positive scaling law would require all seven datasets to have positive slopes with overlapping confidence intervals; we observe none of these.

The same per-dataset fit is overlaid on Figure 1 as a light dashed line per dataset, so the visual mismatch between AwA2's tight log-linear fit and the other datasets' poor or contrary fits is directly inspectable.

The model is actually learning on each dataset. Reporting only the converged c would leave open the alternative explanation that some Pattern C datasets are simply undertrained models whose curvature is collapsing on noise. Table 4 shows zero-shot top-1 and top-5 accuracy on the held-out unseen classes at 100% data, alongside the converged $c_{100\%}$. Every dataset is well above random-chance baseline on top-1 (chance is $1/N_{\text{unseen}}$, never above 2%; the lowest absolute accuracy is tiered-ImageNet at $3.24\% \approx 5\times$ random over its 160 unseen classes). Unseen-class top-5 accuracy is between 12% and 89% across datasets. Notably the relationship between $c_{100\%}$ and accuracy is non-monotonic: AwA2 (highest c) and CUB-200 (mid c) achieve the highest top-1 accuracies (60.8%, 56.6%), while Stanford Cars (mid c) is one of the lowest (11.9%) and SUN397 (lowest c) sits in the middle (26.7%) — so a Pattern C trajectory is not a marker of an undertrained model.

5.2 Three response patterns

The within-dataset trajectories in Figure 1 cluster into three qualitatively distinct shapes. To avoid relying on visual classification we apply the following mechanical 3-way criterion to the curvature trajectory $\{c_{1\%}, c_{5\%}, \ldots, c_{100\%}\}$ of

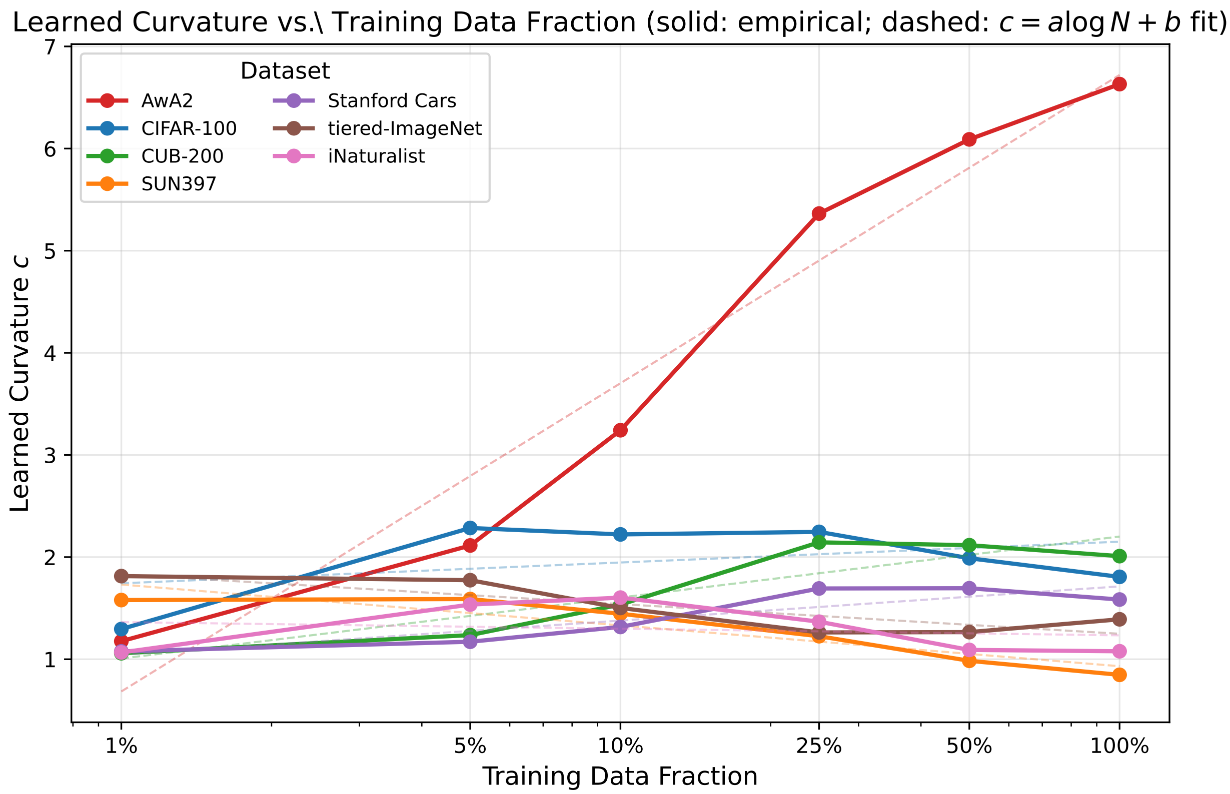

Figure 1: Converged curvature vs. training data fraction (log-x) for all seven datasets. Solid line + markers: empirical trajectory. Light dashed line per dataset: the corresponding least-squares $c = a \log N + b$ fit reported in Table 3. AwA2 (top) climbs almost linearly in $\log N$ and the dashed fit hugs the data. The GloVe-embedded datasets cluster between $c = 0.8$ and $c = 2.3$ and exhibit qualitatively different trajectories: CUB-200 and Stanford Cars rise then plateau; SUN397, iNaturalist, tiered-ImageNet, and CIFAR-100 peak early and decline.

each dataset:

- **Pattern A (monotonic increase)**: $\text{argmax}_f c_f$ falls at the largest fraction (100%) *and* $c_{100\%} > 1.5\, c_{1\%}$.
- **Pattern C (peaks early, declines)**: $\text{argmax}_f c_f$ falls at one of the three smallest fractions ($\leq 10\%$ data) *and* $c_{100\%} < 0.85 \cdot \max_f c_f$.
- **Pattern B (rises then plateau)**: any other configuration (peak in the middle, or peak early but final value within 15% of the peak).

Applied mechanically, the classification is shown in Table 5.

Pattern A — Monotonic increase. AwA2 is the sole member: c rises from 1.18 to 6.63, a 5.6× increase, fitting tightly to a $\log N$ line ($R^2 = 0.95$, slope-CI $[+0.90, +1.72]$). This is the pattern from Bejleri [2025] that motivated the present study.

Pattern B — Rises then plateaus. CUB-200 and Stanford Cars. Both rise initially (positive slope on the early portion of the sweep), reach a moderate peak between $c = 1.7$ and $c = 2.3$ somewhere in the 25–50% data range, and end within 7% of that peak at 100% data. The ceiling is consistent across very different label spaces: CUB-200's deep ornithological hierarchy and Stanford Cars's flat make/model space both produce c around 1.6–2.0 at 100% data, suggesting the ceiling is not driven by the label hierarchy.

Table 4: Zero-shot classification on *unseen* classes at 100% training data, alongside converged curvature $c_{100\%}$. Models are above random across the board (random-chance top-1 is $1/N_{\text{unseen}} \leq 2\%$ for every dataset), confirming that the curvature trajectories in Table 2 reflect actual training dynamics rather than collapse on noise.

Dataset	$c_{100\%}$	Top-1 (%)	Top-5 (%)	N_{test}
AwA2	6.63	60.76	88.69	7 454
CUB-200	2.01	56.65	89.24	2 927
CIFAR-100	1.81	17.32	61.29	12 000
Stanford Cars	1.58	11.85	39.90	2 050
tiered-ImageNet	1.39	3.24	12.08	206 209
iNaturalist	1.08	15.02	40.42	5 000
SUN397	0.85	26.66	58.10	19 499

Table 5: Mechanical pattern classification (criterion in text). Argmax column shows where the per-dataset curvature peaks; the final/max ratio is the diagnostic for Pattern C vs. B.

Dataset	argmax	$c_{1\%}$	$c_{\max}$	$c_{100\%}$	Pattern
AwA2	100%	1.18	6.63	6.63	**A**
CUB-200	25%	1.06	2.14	2.01	B
Stanford Cars	50%	1.08	1.70	1.59	B
CIFAR-100	5%	1.30	2.29	1.81	C
SUN397	5%	1.58	1.59	0.85	C
iNaturalist	10%	1.07	1.60	1.08	C
tiered-ImageNet	1%	1.81	1.81	1.39	C

Pattern C — Peaks then declines. CIFAR-100, SUN397, iNaturalist, and tiered-ImageNet. All four peak at or before 10% of the data and then decline by at least 21% of the peak by 100% data (CIFAR-100: 21%, tiered-ImageNet: 23%, iNaturalist: 33%, SUN397: 47%). SUN397 is the cleanest case: a strict monotonic decline from $c = 1.58$ at 1% data down to $c = 0.85$ at 100% — well below the initial value of $c_0 = 1.0$. We note that the criterion places CIFAR-100 in Pattern C even though its trajectory is qualitatively close to a noisy plateau (peak 2.29, end 1.81); a less-strict threshold would put it on the B/C boundary.

These three patterns describe the data without committing to a mechanism. The next two subsections explore why they appear.

5.3 Embedding modality, not taxonomy depth, predicts the pattern

Figure 2 shows the converged 100%-data curvature against the ground-truth tree depth and branching factor of each dataset's taxonomy. The relationships are weak in both directions. A simple linear regression of $c_{100\%}$ on tree depth across all seven datasets has $R^2 = 0.27$ (driven primarily by AwA2 as a high-depth-3 outlier in the wrong direction), and $R^2 = 0.05$ once AwA2 is excluded. For mean branching factor, $R^2 = 0.02$ (all 7) and $R^2 = 0.13$ (excluding AwA2). Repeating the same analysis for the slope a from Table 3 (rather than the static $c_{100\%}$) gives the same picture: a vs. depth has $R^2 = 0.05$ (all 7), $R^2 = 0.31$ (ex-AwA2); a vs. branching has $R^2 = 0.17$ (all 7), $R^2 = 0.29$ (ex-AwA2). None of these pass even a generous threshold for an explanatory relationship, with or without AwA2.

The single feature that does predict the pattern cleanly is the embedding modality. AwA2 (Pattern A, the only outlier) is the only dataset using **per-class binary attributes** Xian et al. [2019]: 85 expert-annotated dimensions like `has_stripes`, `carnivorous`, `lives_in_water`. These attributes are chosen by domain experts to maximally discriminate animal species, and they encode rich, structured semantic relationships — two animals with identical attribute strings would share a complete biological description. By contrast, the other six datasets use 300-d GloVe word vectors derived from class names (or, for tiered-ImageNet and iNaturalist, from taxonomy-derived words). GloVe vectors are trained on co-occurrence statistics in unstructured text and capture mostly distributional similarity; they do not encode the kind of fine-grained discriminative structure that per-class attributes do.

This distinction predicts:

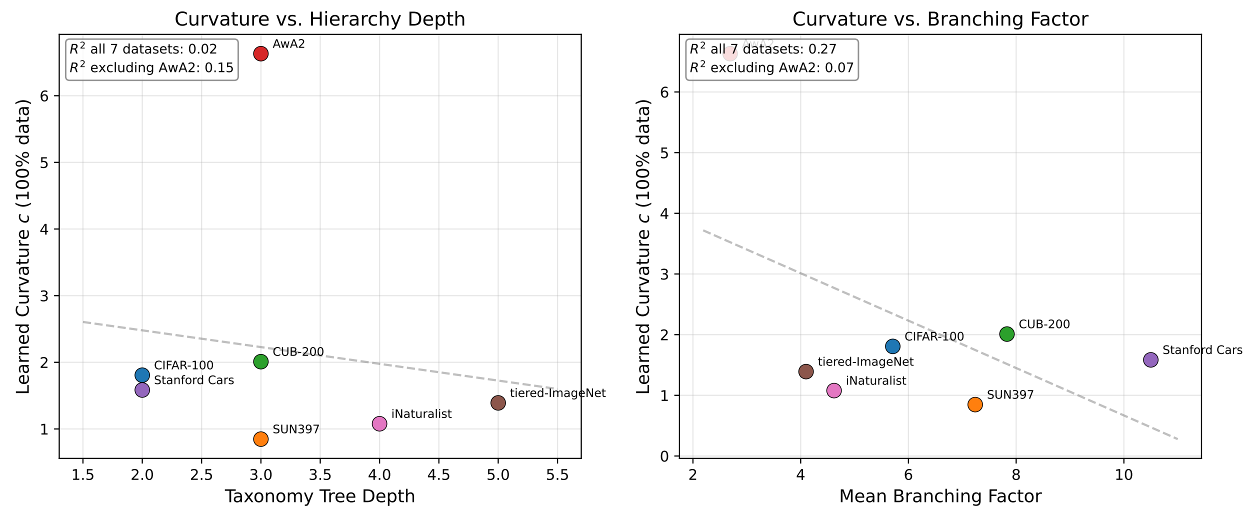

Figure 2: Converged curvature at 100% data vs. (left) taxonomy tree depth and (right) mean branching factor. Each panel reports R^2 both with all seven datasets and with AwA2 excluded; in both panels, removing AwA2 either preserves or reduces the (already weak) correlation, ruling out the reading that AwA2 is creating a spurious signal that the GloVe-embedded datasets would otherwise carry.

- Pattern A appears only when the class embeddings are themselves richly structured. AwA2's curvature climbs because each new training example refines the alignment of visual features with a meaningful, dense per-class semantic description, and the model can keep extracting more geometric structure from the data.
- Patterns B and C appear when the class embeddings are flatter (in the structural sense) than the visual features. The model learns a useful curvature in the small-data regime — when class embedding distances are the only available signal — but as visual features become individually informative (with more data), it discovers that compressing into curved space buys it less than letting the visual encoder do the discrimination directly. This produces the plateau (Pattern B) or, on visually rich datasets with many classes (SUN397, iNaturalist), the active decline (Pattern C).

We do not claim this explanation is the only one consistent with the data. But the magnitude of the AwA2 vs. everything-else gap (a factor of roughly $4\times$) cannot be explained by taxonomy alone, given that CUB-200's depth-3 ornithological hierarchy and CIFAR-100's depth-2 superclass structure both produce curvatures in the same modest range as Stanford Cars's flat make space.

5.4 Number of classes interacts with the pattern

A secondary observation: the four datasets in Pattern C (declining curvature) include the three largest by class count — SUN397 (397), iNaturalist (500), tiered-ImageNet (608) — plus the much smaller CIFAR-100 (100). The two Pattern-B datasets sit at moderate class counts (CUB-200: 200, Stanford Cars: 196). Regressing the converged $c_{100\%}$ on $\log(\text{n\_classes})$ across all seven datasets gives $R^2 = 0.65$, and on the six GloVe-embedded datasets alone (i.e., controlling for embedding modality) $R^2 = 0.51$ with a negative slope — more classes, lower converged curvature. Repeating for the slope a: $R^2 = 0.70$ across all seven, $R^2 = 0.50$ excluding AwA2. Both relationships survive removing AwA2, which suggests class count is a real — though noisy and clearly not the only — factor.

A plausible reading is that with hundreds of classes embedded into 300-d GloVe space, many class embeddings end up near each other (especially fine-grained taxonomic classes whose names share roots), and large-curvature embeddings exaggerate small differences as noise rather than signal. The model then unlearns its initial curvature to compensate. This explanation is consistent with tiered-ImageNet (608 classes drawn from WordNet leaves whose lemmas often duplicate or differ by a single qualifier) showing the lowest peak curvature among the GloVe-embedded large datasets and the steepest early decline. We caution that with only seven data points, the regressions reported here are diagnostic rather than confirmatory, and CIFAR-100 (only 100 classes but firmly Pattern C) is a clear counterexample to a strict "more classes $\Rightarrow$ Pattern C" reading.

5.5 Comparison with Fan et al.

Fan et al. Fan et al. [2025] report converged curvature values from a sharpness-aware bi-level optimisation procedure on a few-shot classification objective. Their reported $c \approx 0.113$ on tiered-ImageNet 1-shot is roughly an order of magnitude smaller than our $c = 1.26$–1.81 on the same dataset. Before reading anything into this gap, we flag a parameterisation question that has to be settled first. The hyperbolic-network literature is not consistent about what "c" denotes: `geoopt` Kochurov et al. [2020] (which we use) defines the Poincaré ball with constant negative sectional curvature $-c$ for $c > 0$ via the conformal factor $\lambda_x^c = 2/(1 - c\|x\|^2)$ on the unit ball, while several other works use the radius $r = 1/\sqrt{c}$, the curvature itself $\kappa = -c$, or a dimensionless $\sqrt{c}$-scaled coordinate. A $10\times$ apparent gap in c-values across papers can correspond to a $\sqrt{10} \approx 3.2\times$ difference in radius or to using the same geometry under a different convention entirely. Without re-deriving Fan et al.'s parameterisation from their code we cannot rule out that some or all of the order-of-magnitude gap is convention; we therefore restrict our comparison to qualitative claims.

What is meaningful at this resolution is that both papers find tiered-ImageNet's optimal/converged curvature in a low-to-moderate range that does not grow without bound with more data, consistent with the more general observation that text-derived class embeddings on large class sets produce only moderate curvature — in striking contrast to AwA2's attribute-driven climb to $c = 6.63$ in our parameterisation. Quantitative cross-paper comparisons of c should be done either after explicit unit normalisation (e.g., reporting $\sqrt{c} \cdot$ embed-dim or the geodesic ball radius) or by reproducing both methods under the same library.

A more important point: our results suggest that "optimal curvature for tiered-ImageNet" is best understood as "optimal curvature for tiered-ImageNet *when classes are encoded by GloVe word vectors*." A different class encoding — per-class attributes, learned visual prototypes, language-model embeddings — would likely produce a different optimum. Prescriptive curvature-tuning procedures should report (and ideally vary) the embedding modality alongside the dataset.

6 Discussion

Curvature reports semantic structure as the model can see it. The picture that emerges is not "curvature reports the dataset's hierarchy depth." It is "curvature reports how much hierarchical/relational structure the model can extract from its inputs given how the classes are encoded." On AwA2, that structure is rich and the model keeps finding more of it as data grows. On GloVe-embedded datasets, the structure available through 300-d word vectors is shallower, and the model converges to a moderate curvature ceiling — or, when the class set is large enough that GloVe distances stop being discriminative, retreats from curvature entirely.

Implications for hyperbolic network design. A practical consequence: when designing a hyperbolic network for a new task, the choice of class embedding may matter as much as the choice to use hyperbolic geometry. A hyperbolic network with weak class embeddings may end up learning a near-Euclidean curvature in any case, defeating the purpose of the architectural choice. Practitioners might reasonably treat the converged curvature as a diagnostic: if c stays near c_0 or drops below it, the embedding likely lacks the structure the geometry was meant to exploit, and either the embedding or the architecture should be reconsidered.

Hypothesis: why does AwA2 keep climbing? The c values for AwA2 do not plateau within our 100% data sweep, raising the question of whether they would saturate at higher data — noting that AwA2 has only $\sim$30K images, so we cannot push much further on this dataset alone. We offer the following as a *hypothesis*, not a derivation: AwA2's 85 expert-annotated binary attributes provide a class-conditional supervisory signal that varies meaningfully between training examples of *different* classes (the attribute vector for "tiger" differs from "polar bear" in many dimensions, each one a learnable axis of the visual–semantic alignment). GloVe word vectors, by contrast, provide a fixed 300-d distributional signal per class that is largely redundant across examples of the same class and varies only along distributional-similarity directions across classes, regardless of how many examples are seen. On this account, AwA2's curvature climbs because each additional training example refines the model's alignment of visual features with semantically meaningful attribute axes, while GloVe-embedded models hit a ceiling once the embedding's distributional structure has been absorbed. The class-attribute side, not the image side, is what we expect to determine the saturation point: an AwA2-style architecture trained with longer natural-language descriptions or visual-language model embeddings might produce even steeper Pattern A curves, while AwA2 trained with GloVe class-name embeddings would (we predict) drop to Pattern B or C. This hypothesis is testable; we mark the GloVe-AwA2 control as future work in Section 6.

What about δ-hyperbolicity? We computed taxonomy-derived hierarchy metrics (depth, branching factor) but did not include feature-space δ-hyperbolicity Gromov [1987], Fournier et al. [2015] in the main analysis. A natural extension is to compute δ on the joint visual-feature plus class-embedding metric and see whether it correlates with the converged curvature better than taxonomy depth does — particularly since δ-hyperbolicity is sensitive to the embedding metric in a way that taxonomy depth is not.

Limitations. We list the major confounds and what controlled comparison (existing or proposed) speaks to each.

(L1) Single seed. Every number in Tables 2, 3 and 5 and every trajectory in Figure 1 comes from a single training run per (dataset, fraction) cell. The 95% CIs in Table 3 reflect within-trajectory linear-regression uncertainty, not cross-seed variance. Multi-seed reruns are needed before any quantitative claim about slope magnitude or about the B/C boundary should be considered confirmed; the qualitative claims — AwA2 climbs to $c \approx 6.6$, SUN397 declines from $c_0 = 1.0$ to $c = 0.85$, and these two cannot share a single $c \propto \log N$ scaling law — are large enough to survive plausible single-seed noise, but the fine-grained distinctions cannot.

(L2) AwA2 is single-anchor on six of the dimensions that distinguish it from the other datasets. AwA2 differs from the other six in: (a) embedding modality (per-class binary attributes vs. GloVe word vectors), (b) embedding dimensionality (85 vs. 300), (c) class count (50, the smallest), (d) domain (animals only, very narrow), (e) dataset size ($\sim$30K images, on the small end), and (f) hierarchy depth among datasets with < 200 classes (depth 3 vs. CIFAR-100's depth 2). Our embedding-modality hypothesis is the cleanest single-factor explanation for the AwA2 vs. everything-else gap, but it is not the only one consistent with the data. Within-existing-data we can rule out (c) as the dominant factor because the regression of $c_{100\%}$ on $\log(\text{n\_classes})$ across the six GloVe-only datasets has $R^2 = 0.51$ — real but far from determining — and we can rule out (f) because CIFAR-100 (depth 2) and CUB-200 (depth 3) have very similar converged curvatures. We cannot disentangle (a), (b), (d), and (e) from the existing six-dataset GloVe set: AwA2 is the only attribute-embedded, animals-only, $\sim$30K-image dataset. A clean test for (a) would re-run the same architecture on AwA2 with 300-d GloVe class-name embeddings; if the trajectory drops to Pattern B or C, (a) is the active factor. We mark this as future work.

(L3) Hierarchy depths are conservatively truncated. Our `hierarchy.py` truncates iNaturalist (full depth 7) and tiered-ImageNet (full WordNet depth $\approx$12–15) to coarsely meaningful levels. The truncation compresses the depth axis but preserves ordering. The negative result on depth-vs-curvature would only strengthen if the un-truncated depths were used.

(L4) Initial curvature is fixed at $c_0 = 1.0$. Pattern C, especially SUN397's decline to $c = 0.85$, partly looks like "the model unlearned its initial curvature." We have not verified that converged $c_{100\%}$ is initialisation-invariant on Pattern C datasets. A small ablation varying $c_0 \in \{0.1, 1.0, 5.0\}$ on SUN397 would show whether the patterns are real attractors or just slow drift from c_0.

7 Conclusion

We tested whether the $c \propto \log N$ scaling of learned Poincaré ball curvature observed on AwA2 generalises across image classification datasets. It does not. Across seven datasets, the within-dataset slope of c vs. $\log N$ varies from $+1.31$ (AwA2) to -0.17 (SUN397), and the trajectories cluster into three distinct response patterns. The strongest predictor of which pattern a dataset exhibits is the type of class embedding used — per-class attributes (AwA2) versus word vectors (everything else) — and not the depth of the underlying taxonomy. We re-frame the descriptive question accordingly: learned curvature reports the dataset's semantic structure *as visible through its embedding modality*, not its taxonomy.

A Training-loss and curvature convergence at low data fractions

A natural concern with the Pattern C datasets is that an early-fraction curvature peak — e.g. tiered-ImageNet at $c = 1.81$ for 1% data — could just be the optimiser failing to reach steady state with very little data, with c frozen near its initialisation rather than reflecting a converged optimum. Figure 3 plots the training loss (top row) and the learned curvature (bottom row) over the full 200-epoch training schedule for every dataset at the 1% and 10% ablation points.

In all cases the training loss is monotonically decreasing and visibly plateaus by the end of training, and the curvature parameter is also stable at the end of training (i.e., the slope of c vs. epoch in the final $\sim$50 epochs is close to zero), confirming that the converged values reported in Table 2 are not artifacts of an unconverged optimiser at low-data fractions.

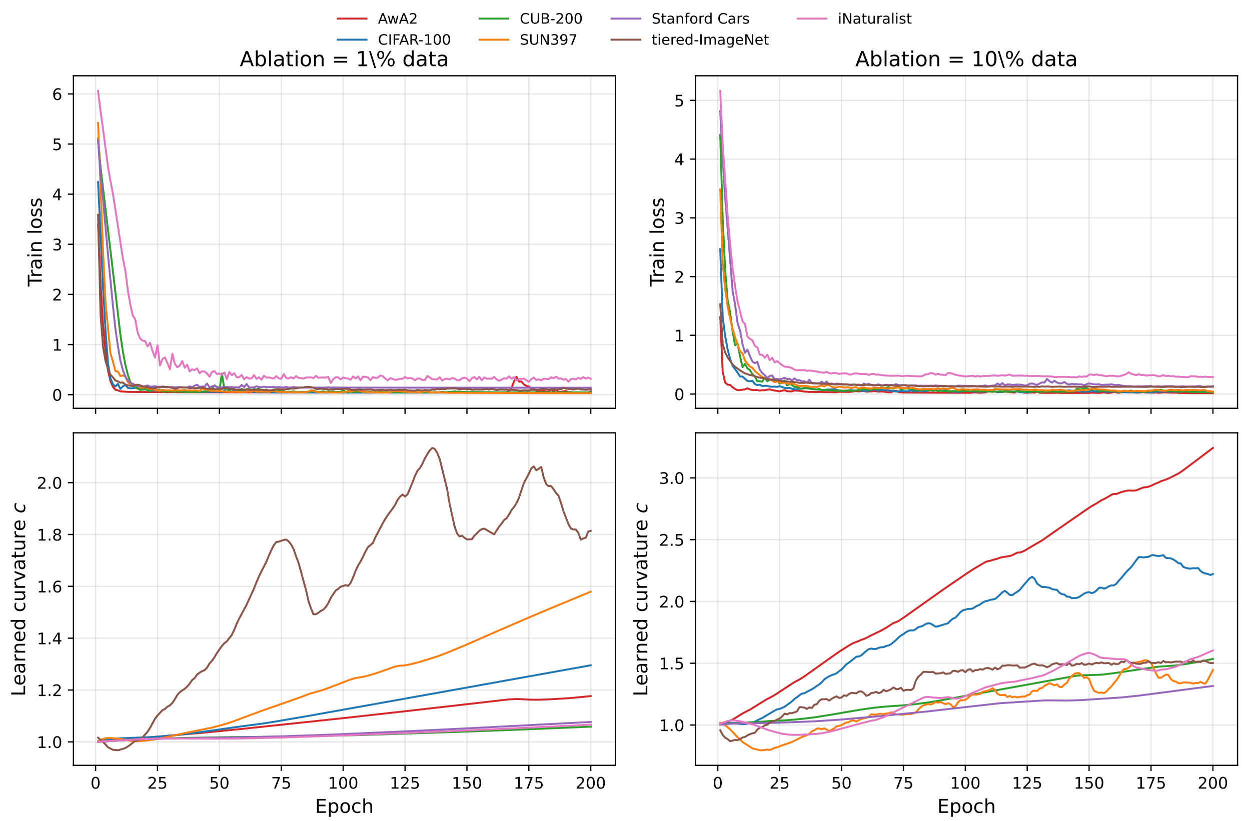

Figure 3: Training loss (top) and learned curvature c (bottom) over 200 training epochs for every dataset, at the 1% (left column) and 10% (right column) ablation points. Training loss decreases monotonically and the curvature parameter stabilises in all cases, ruling out the explanation that early-fraction curvature peaks reflect an unconverged optimiser.

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